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William Mitchell, Scott Viteri, Mehmet Tugrul Savran
{williamm, sviteri, tugrul}@mit.edu

6.863: Natural Language Processing

Synonym Finder

1. Problem Statement

Synonym generation from a corpus has been a prevalent domain of research in Natural Language Processing. The synonym finding problem is to find the most appropriate synonym for a word that will help the sentence preserve its original meaning; the model chooses the closest word from a given set of options.

An example of the Synonym Finding problem is realized below:

The sentence:

The President ____ with the Vice President, and exchanged ideas about the status quo.

Choices:

- A) Cat
- B) Glanced
- C) Met
- D) NLP

Accordingly, the computational pipeline breaks down into two major components. The first is the *generation* of candidate words such as “Cat” or “Glanced” above. This process is analogous to enumerating or coming up with the “choices” in a multiple choice exam. The second process is to *iterate* through the set of choices and pick the best option.

Here, the computational model should test each of the 4 possible choices, and pick the most appropriate option, C.

The rest of the proposal continues with the discussion of methodologies for the computational pipeline. Namely, we will experiment with different approaches for generating candidate synonyms, and test various methods that will strive to pick the best synonym option.

Section 2 continues with such discussion.

2. Methods

2.1 Sentence Generation

As Section 1 indicated, the first part in the pipeline is to generate sentences and also options the computational model will choose from. Having an automated sentence and choice generator will be crucial for testing the models. Hence we plan on using our English grammars from lab 2 to generate sentences. Then we select a random word and remove it. We provide a list of options to replace the removed word. We provide one synonym by using a github front-end to wordnet (<https://github.com/rm-hull/clj-wordnet>) using the command (map :lemma (words (:synset dog))), making sure that the selected synonym is in the training set. For the rest of the example words, we pick a bunch of random words from the word2vec training set as well.

2.2 Synonym Finding Model

The “brain” of the pipeline will be the synonym finding model. We will train a vector representation of words using some corpus of sentences. We will compare and contrast several different methods, including continuous bag-of-words, skip-gram, and Poincare embeddings (<https://arxiv.org/pdf/1705.08039.pdf>). For each of these examples, we will look at different distance metrics between vectors -- cosine, inner product, L2, L1 distance, and L-infinity norm.

For each vector embedding and each distance metric, we will show examples of situations that work, and situations that do not. For tests that do not consistently pick the synonym, we will attempt to understand the reason why the prediction failed, and test those hypotheses.

Next, we desire to come up with a metric for how well each variant functions and test how well each model/metric performs. For the initial testing stage, we will use a set of human-produced examples. Next, for particularly successful models, we will utilize the automated sentence/choice generator mentioned above in Section 2.1.

3. Goals/Extensions

We have proposed exploring three different methods for synonym matching problems: by using continuous bag of word, n-gram, and Poincare embeddings. Our goal is to provide examples for where each succeeds or fails and to use the various distance metrics -- cosine, inner product, L2, L1 distance, and L-infinity norm -- to judge which method is best. In the context of

the problem, the determination of which method is best will be measured by consistency in producing valid synonyms. We seek to determine whether one method regularly outperforms the others. If we find that none of them are better than the others with any consistency, that in itself will be an interesting result.

After looking at the feasibility of the previously mentioned methods for determining valid synonyms, the next frontier will be to determine context of whole sentences. This would potentially include the construction of sentences that are, for all intents and purposes, equivalent in their meaning but made up of different sets of words. We could also explore context of sentences that belong to bodies of text.

4. Relevant Literature

Word2Vec:

<https://papers.nips.cc/paper/5021-distributed-representations-of-words-and-phrases-and-their-compositionality.pdf>

Poincare Embeddings: <https://arxiv.org/pdf/1705.08039.pdf>

Wordnet: <http://wordnetcode.princeton.edu/5papers.pdf>